RAY OPTICS

FORMULA SHEET | JEE MAINS | Let's Play With Physics

1. SIGN CONVENTION & PLANE MIRROR

Cartesian convention: distances from pole/optical centre; along incident light = +, against = −; heights up = +.

Deviation by plane mirror: $\delta = 180^\circ - 2i$
Mirror rotated by $\theta \Rightarrow$ reflected ray rotates by 2θ
Min. mirror length for full image $= \dfrac{h}{2}$

Images between two mirrors at angle θ: let $m = \dfrac{360^\circ}{\theta}$

- m even: $n = m - 1$
- m odd: $n = m - 1$ (object on bisector), $n = m$ (otherwise)

Image velocity (plane mirror): $\vec{v}_{I\perp} = 2\vec{v}_{M\perp} - \vec{v}_{O\perp}$, parallel components equal

⋆ Common Pitfall

- Plane mirror image: virtual, erect, same size, laterally inverted
- Only the velocity component **normal** to the mirror reverses

2. SPHERICAL MIRRORS

$$f = \frac{R}{2}, \qquad \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$m = \frac{h_i}{h_o} = -\frac{v}{u} = \frac{f}{f-u} = \frac{f-v}{f}$$

Longitudinal magnification: $m_L = \dfrac{dv}{du} = -\dfrac{v^2}{u^2} = -m^2$
Image speed along axis: $v_I = -m^2\, v_O$
Newton's formula: $x_1 x_2 = f^2$ (distances from focus)

Object (concave)	Image at	Nature
∞	F	real, inverted, point
Beyond C	F to C	real, inverted, diminished
At C	C	real, inverted, same size
C to F	Beyond C	real, inverted, enlarged
At F	∞	real, inverted, highly enlarged
F to P	Behind mirror	virtual, erect, enlarged

Convex mirror: always virtual, erect, diminished, between P and F.

⋆ Common Pitfall

- Concave: $f < 0$; convex: $f > 0$ (real object has $u < 0$)
- $f = R/2$ holds only for paraxial rays
- Focal length of a mirror does not depend on the surrounding medium

3. REFRACTION, SLAB & APPARENT DEPTH

$$n_1 \sin i = n_2 \sin r, \qquad n = \frac{c}{v}, \qquad \lambda_{\text{med}} = \frac{\lambda_0}{n}$$

Frequency does not change on refraction.

Apparent depth (near-normal view): $h' = \dfrac{h}{n}$

Normal shift by slab: $\Delta = t\left(1 - \dfrac{1}{n}\right)$ (towards observer)

Layers: $h' = \sum \dfrac{t_i}{n_i}$

Lateral shift: $d = \dfrac{t \sin(i-r)}{\cos r}$

⋆ Common Pitfall

- Slab shift $t(1 - 1/n)$ is independent of object distance
- Emergent ray from a slab is parallel to incident ray

4. TOTAL INTERNAL REFLECTION

$$\sin C = \frac{n_2}{n_1} = \frac{1}{n} \quad \text{(to air)}$$

Conditions: denser → rarer, and $i > C$

Medium	n	C
Water	4/3	48.6°
Glass	1.5	41.8°
Diamond	2.42	24.4°

Source at depth h: radius of bright circle $r = h\tan C = \dfrac{h}{\sqrt{n^2-1}}$

Optical fibre: NA $= \sin\theta_{\max} = \sqrt{n_1^2 - n_2^2}$ (core n_1, cladding n_2, in air)

⋆ Common Pitfall

- TIR is impossible from rarer to denser medium
- Uses: optical fibre, totally reflecting prisms, mirage, sparkle of diamond

5. REFRACTION AT SPHERICAL SURFACE

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}, \qquad m = \frac{n_1\, v}{n_2\, u}$$

Plane surface ($R \to \infty$): $v = \dfrac{n_2}{n_1}\, u$

⋆ Common Pitfall

- n_1 = medium of the object (incident side), n_2 = refracted side
- $R > 0$ if centre of curvature lies on the refracted-light side

6. THIN LENSES

$$\frac{1}{f} = \left(\frac{n_\ell}{n_m} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}, \qquad m = \frac{v}{u}, \qquad P = \frac{1}{f\,(\text{m})}\ \text{D}$$

Lens in a liquid: $\dfrac{f_{\text{liq}}}{f_{\text{air}}} = \dfrac{n_g - 1}{n_g/n_l - 1}$

- $n_l > n_g$: convex lens behaves as diverging
- $n_l = n_g$: lens becomes invisible ($f \to \infty$)

Combination: in contact $\dfrac{1}{F} = \sum \dfrac{1}{f_i}$, $P = \sum P_i$

Separated by d: $\dfrac{1}{F} = \dfrac{1}{f_1} + \dfrac{1}{f_2} - \dfrac{d}{f_1 f_2}$

Silvered lens (acts as a mirror): $P_{\text{eq}} = 2P_{\text{lens}} + P_{\text{mirror}}$

- Plano-convex, plane face silvered: $f_{\text{eq}} = \dfrac{R}{2(n-1)}$
- Plano-convex, curved face silvered: $f_{\text{eq}} = \dfrac{R}{2n}$

Displacement method: object–screen distance D, lens positions x apart:

$$f = \frac{D^2 - x^2}{4D}, \qquad D \geq 4f, \qquad h_o = \sqrt{h_1 h_2}$$

Cutting a lens: along the axis ⇒ each half has same f (less intensity); equiconvex cut ⊥ axis into two plano-convex ⇒ each $f' = 2f$

⋆ Common Pitfall

- Min. distance between real object and its real image (convex lens) $= 4f$
- In $1/f$ formula use $n_{\text{lens}}/n_{\text{medium}}$, not n_{lens}
- Convex lens: $f > 0$, $P > 0$; concave: $f < 0$

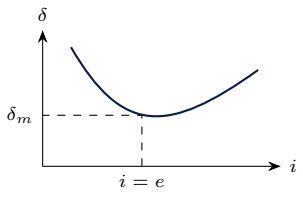
7. PRISM

$$r_1 + r_2 = A, \qquad \delta = i + e - A$$

Minimum deviation ($i = e$, $r_1 = r_2 = A/2$):

$$n = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Thin prism: $\delta = (n-1)A$



No emergence for any i if $A > 2C$.

⋆ Common Pitfall

- At δ_m the ray passes symmetrically (parallel to base for isosceles prism)
- Two values of i give the same δ (except at δ_m): i and e interchange

8. DISPERSION

Angular dispersion: $\theta = \delta_v - \delta_r = (n_v - n_r)A$

Mean deviation: $\delta_y = (n_y - 1)A$

Dispersive power: $\omega = \dfrac{n_v - n_r}{n_y - 1} = \dfrac{\theta}{\delta_y}$

Achromatic (no dispersion): $\omega\delta + \omega'\delta' = 0$

Direct vision (no deviation): $(n_y - 1)A + (n'_y - 1)A' = 0$

Achromatic lens doublet: $\dfrac{\omega_1}{f_1} + \dfrac{\omega_2}{f_2} = 0$

Cauchy: $n = a + \dfrac{b}{\lambda^2} \Rightarrow n_v > n_r$, violet deviates most

⋆ Common Pitfall

- Dispersive power is a property of the material, independent of A
- In vacuum there is no dispersion (all colours travel at c)

9. OPTICAL INSTRUMENTS

Simple microscope ($D = 25$ cm):

$$M = \frac{D}{f}\ \text{(image at } \infty\text{)}, \qquad M = 1 + \frac{D}{f}\ \text{(image at } D\text{)}$$

Compound microscope: $M = m_o\, m_e$

$$M_\infty = \frac{v_o}{|u_o|} \cdot \frac{D}{f_e} \approx \frac{L}{f_o} \cdot \frac{D}{f_e}$$

$$M_D = \frac{v_o}{|u_o|}\left(1 + \frac{D}{f_e}\right)$$

Astronomical telescope:

Normal adjustment: $M = \dfrac{f_o}{f_e}$, length $L = f_o + f_e$

Final image at D: $M = \dfrac{f_o}{f_e}\left(1 + \dfrac{f_e}{D}\right)$, $L = f_o + |u_e|$

⋆ Common Pitfall

- Microscope: small f_o & small f_e; telescope: large f_o, small f_e
- Reflecting telescope: no chromatic aberration, easier mechanical support for large aperture
- Max magnification when final image is at near point D

10. JEE SHORTCUTS & MASTER TIPS

1. Mirror: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$; Lens: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
2. Mirror $m = -v/u$; Lens $m = +v/u$
3. Slab shift $= t(1 - 1/n)$, independent of position
4. f_{eq} silvered lens: $P = 2P_L + P_M$
5. Thin prism: $\delta = (n-1)A$; lens in contact: $P = \sum P_i$
6. Displacement method: $f = (D^2 - x^2)/4D$

⋆ Master Confusion List

1. **Sign of f:** concave mirror −, convex lens +
2. **Medium:** mirror f unchanged in water; lens f changes
3. $m_L = -m^2$ only for small objects along the axis
4. **Refraction formula:** n_1 is object side, not the denser side
5. **TIR:** only from denser to rarer, with $i > C$
6. **Telescope length:** $f_o + f_e$ only in normal adjustment

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